

create a rules database. In addition to creating the rule book, the framework will need to include a monitoring application to continuously monitor incoming data, and generate alerts and reports whenever the incoming data matches a rule in the rule database.

Pattern discovery to solve business problems

FPM algorithms can be used to provide solutions for various domains. In this section, I will describe some of the solutions built using the concept of pattern discovery, where these algorithms were applied. The solutions cover multiple domains. In each of the solutions, various algorithms are leveraged to mine patterns from the data, which can be used as per the need of the application.

Market-basket analysis: This solution was developed to discover purchasing patterns from the customers' shopping cart data. Patterns were discovered using associations or by identification of co-occurrences from transactional data associated with customers' shopping carts.

Market-basket analysis techniques were used to mine the consumer purchase patterns. This technique allows retailers to consider the customers' market basket, in order to understand which products are purchased together, the frequency of purchases as well as the volume of purchases. Such a mining technique enables retailers to formulate an effective strategy for campaigns and promotions. This, in turn, results in increasing the customer base as well as increasing the value of the market-basket.

Machine log analytics: One of our manufacturing customers, wanted to implement an enterprise application to monitor software patches for the company's medical devices across the world. The need to install a software patch was determined by the sequence of codes that the device generated during its operation.

While the SME provided some sequences of frequently occurring codes, this manual process of pattern identification was quite a time consuming and effort-intensive activity. To save time and effort, we applied frequent item-set mining algorithms to identify sequences. The sequences generated by the algorithm were validated by the SME. Then the patterns were converted into business rules. The rules were deployed to identify the relevant software patch for a specific device. Due to the number of devices deployed and their geographical distribution, the earlier, manual process of identifying patterns was time-consuming. The automated process of pattern identification acted as an enabler to improve productivity.

Fraud analytics: With the Internet being widely accessible, customers have adopted e-commerce for all their purchases. This development goes hand-in-hand with the use of credit and debit cards for making

payments. In an e-commerce transaction, a person is not physically present for verification and it is possible to make a purchase simply with a card number. Many customers these days are facing various attacks like phishing, pharming, skimming and dumpster driving. Hence, there is a need to protect against such attacks.

To address this business need, we built a model of customer behaviour. This model is built on an unsupervised learning method. Every customer transaction is validated against the customer's profile to identify anomalies, if any. If the anomaly check fails, alerts are generated. A tree-based pattern mining algorithm can be leveraged to solve this problem. This algorithm enables identification of the transaction patterns which typically trigger fraudulent behaviour.

Future work

Gartner predicts that, in a few years, smart pattern discovery will be in huge demand, enabling mainstream business consumers to get insights from data. These innovations have the potential to help enterprises gain access to sophisticated interactive analysis and insights.

FPM techniques can be leveraged effectively to discover patterns by finding frequent transaction sets and uncovering the hidden information.

While FPM techniques have a deep impact on businesses through smart pattern discovery, there are still some challenging research issues in terms of selecting the right variation of the techniques to address a specific business problem. 

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